

When Does Network Diversity Reflect Physical Diversity?

A Cross-Layer Audit of Internet Service Paths

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Abstract—Assessing whether network-layer path diversity reflects physical diversity requires comparing the two layers over the same observations. We present CLASP, a Cross-Layer Audit of Service Paths that compares network-transition and feasible landing-region-corridor distributions over the same traceroute segments, with each segment contributing equal total mass at both layers. We apply CLASP to 490,911 valid IPv4 traceroutes from 18 public RIPE Atlas measurements covering all 13 DNS Roots, three applications, and two dynamic multi-target topology references, yielding 370 auditable country–measurement units. Within units, broad network support can coexist with concentrated corridor support: under equal-share allocation, 81 of 275 service-facing units exhibit this combination, while 158 remain broad at both layers. Effective category count contracts in 204 units (74.2%). Across units, the Spearman association between network and corridor concentration, termed Layer Agreement, varies by geography. Under projection-score weighting it is 0.632 for 67 island and archipelagic units and 0.001 for 202 coastal-mainland units. The island–coastal difference remains positive under equal-share, projection-score, and Top-1 allocation, at 0.461, 0.631, and 0.715. These results show why auditing physical-corridor diversity requires both measuring within-unit cross-layer differences and examining how concentration rankings correspond across geographic populations.

Index Terms—Internet measurement, submarine cables, traceroute, cross-layer analysis, anycast, path diversity

I. INTRODUCTION

Submarine cables form the physical backbone of the global Internet and carry nearly all intercontinental traffic [1]. For users accessing services across oceans, these cables provide the physical connections underlying end-to-end network paths. Traceroute exposes the network-layer structure of these paths through visible routers and Autonomous System (AS) transitions between sources and service destinations.

Failures in this physical infrastructure can isolate regions, degrade remote-service access, and redirect traffic toward longer or congested alternatives. Paths traversing different routers or ASes may nevertheless share a cable or coastal corridor and be affected by the same physical event. Such shared dependencies are not directly visible in traceroute: a service may present many distinct network transitions while its feasible physical support remains concentrated in a few coastal directions. Assessing service-path diversity therefore requires comparing how observations are distributed across network transitions and feasible physical corridors.

Cross-layer mapping makes the physical support of network paths accessible to measurement. Internet Atlas, InterTubes, and iGDB connect observed links to infrastructure, while

Nautilus and Calypso infer candidate submarine connections; Xaminer uses cross-layer maps to analyze infrastructure risk [2]–[7]. These capabilities provide the basis for a further comparison: whether the diversity visible across a population of service paths corresponds to the distribution of its feasible physical support. This comparison concerns the relationship between two layers, rather than the physical assignment of an individual path alone.

Comparing network transitions with feasible corridors requires more than counting distinct categories. The two layers describe different categories, and their counts do not reveal whether observation mass is dispersed or repeatedly concentrated in a few categories. A segment may also admit several physical candidates, so its contribution must be allocated without counting each candidate as another observed segment. Finally, the two distributions must describe the same source–measurement population and the same candidate-bearing segments. Otherwise, differences in the sampled paths can be mistaken for differences between layers.

We present *CLASP*, a Cross-Layer Audit of Service Paths, to address these issues through paired distributions. CLASP extracts atomic segments from traceroutes, records their network-transition labels, and constructs feasible landing-region corridors using geographic, cable-lifecycle, and propagation constraints. Each candidate-bearing segment contributes one unit of observation mass at each layer; physical-layer allocation divides that mass among its feasible corridors. Aggregating the same segments within each country–measurement unit then retains both their network identities and their physical support. The resulting distributions allow concentration and breadth to be compared without changing the observation population between layers.

The audit separates two questions that a single diversity summary cannot answer together. **RQ1** asks how concentration and breadth differ between layers within a country–measurement unit. **RQ2** asks whether concentration rankings correspond across units and how this association varies by geography and candidate allocation. We call this Spearman rank association *Layer Agreement*. Aligned rankings do not imply equal breadth within each unit, so the two comparisons are reported together. **RQ3** examines exposure, absolute corridor breadth, and measurement-family composition to contextualize the cross-layer results.

We apply CLASP to 490,911 valid IPv4 traceroutes from 18 public RIPE Atlas measurements covering all 13 DNS Roots,

Wikipedia, Reddit, Netflix Assets, and two dynamic multi-target topology references. The aligned one-hour snapshot yields 370 auditable country-measurement units. Retaining each measurement’s selected targets supports service-specific comparisons, while the multi-target measurements provide a broader-target reference. Equal-share, projection-score, and Top-1 allocation examine the results under alternative distributions of mass within the feasible candidate sets.

The measurements show broad network support alongside concentrated corridor support in some service-facing populations, with additional narrowing among populations classified as broad at both layers. They also show that concentration rankings track more closely among island and archipelagic units than among coastal-mainland units under all three allocation rules. These findings support examining physical-corridor distributions alongside network-layer summaries rather than treating the two as interchangeable descriptions of service-path diversity.

This paper makes three contributions. First, CLASP constructs paired network-transition and feasible-corridor distributions over identical traceroute segments, separating trace-level exposure from conditional corridor concentration. Second, the audit measures within-unit diversity differences and geographic variation in concentration-rank association, with three candidate-allocation rules providing complementary views of the feasible support. Third, it records measurement IDs, configuration values, mapping states, and pipeline counts so that the reported distributions can be reconstructed and inspected.

II. BACKGROUND AND RELATED WORK

A. Reading Service Paths at the Network Layer

A service-path population is defined by both its sources and its measured targets. DNS resolution, replica selection, and routing determine the instances and paths observed from those sources. Traceroute exposes visible IP hops, from which AS transitions can be constructed. A population aimed at a selected service and one aimed at many changing destinations therefore need not describe the same network footprint. Source and measurement identity must remain explicit when comparing their diversity.

Measurements of user-facing infrastructure dependence retain this context in different ways. Liu et al. identify popular Web resources for each country, measure their dependence on the submarine cable network, and consider redundancy using cable and bundle counts [8]. Anycast studies characterize service deployment and network footprints [9], [10]; other work relates cable infrastructure to AS-level structure [11]. These perspectives motivate retaining the service, source, and physical-support context rather than describing diversity through network labels alone.

The physical object used in this study is a *landing-region corridor*, a direction-independent pair of bounded coastal regions connected by at least one feasible cable candidate. Exact landing stations distinguish nearby facilities, whereas country-level inventories combine separated coastlines; landing regions retain coastal structure between these scales. Several cables

may support one corridor, and a network transition may admit more than one corridor. These identities describe different resolutions of feasible support, so neither cable multiplicity nor corridor multiplicity alone establishes how support is distributed across the observed paths.

B. Mapping Paths onto Submarine Infrastructure

Studies of infrastructure change establish why observed routing and physical connectivity should be examined together. Bischof et al. characterized Cuba’s connectivity before and after ALBA-1 and later formulated the task of connecting Internet observations to the worldwide cable mesh [12], [13]. Fanou et al. measured routing changes following new cable deployments [14]. Complementing these event-oriented studies, Liu et al. combine resource discovery, RIPE Atlas traceroutes, geolocation, and propagation feasibility to identify submarine dependence in user access [8].

Cross-layer cartography extends this connection through explicit infrastructure mappings. Internet Atlas and InterTubes associate long-haul facilities with observed paths [2], [3], and iGDB integrates facilities, fiber, and network entities into a cross-layer map [4]. Nautilus associates IP links with candidate submarine cables and attaches confidence to each assignment [5]. Calypso combines cable layouts, network relationships, latency, and inland connectivity to map routes and defines Route Stress over those assignments [6]. Xaminer uses cross-layer maps for disaster-oriented infrastructure analysis [7]. Measurements of intercontinental links also identify layer-3 links and gateway routers [15].

These approaches already support analyses beyond individual mappings, including resource dependence, cable importance, and infrastructure risk. Our study uses physical projection to compare two distributions over identical observations. It asks whether concentration among network transitions corresponds to concentration among feasible corridors, within units and in their ordering across units. Feasible candidate sets are retained throughout this comparison; the physical distribution represents candidate support rather than confirmed traffic shares on specific cables.

C. From Mapping to Distribution Measurement

Aggregating observation mass across feasible corridors measures how repeatedly the observed segments support the same corridors. Two populations can admit the same number of corridors while differing in how much observation mass supports each one. If candidate sets repeatedly overlap, many network transitions may support the same few physical directions. Counting distinct transitions or corridors alone cannot distinguish that organization from more dispersed support. Aggregating the observations at each layer provides the concentration and breadth measures needed for the comparison.

Exposure and concentration answer different parts of the dependence question. *Submarine exposure* records the fraction of valid traceroutes containing at least one feasible inter-region corridor. *Corridor concentration* describes how observation

mass is allocated among corridors, conditional on candidate-bearing segments. Frequent exposure can accompany either broad or concentrated support. Reporting the two quantities separately preserves the distinction between how often paths enter the feasible corridor space and how support is distributed within that space.

The unit of comparison must preserve the service context introduced above. CLASP groups observations by source country and measurement: DNS Roots retain their independent anycast deployments, the three applications retain their selected targets, and the two dynamic multi-target measurements provide broader-target topology references. The latter are reference populations, not matched controls for service deployment. This grouping keeps source and target context explicit when interpreting differences among units.

Within each unit, the comparison must also preserve observation mass. Each candidate-bearing segment has one network-transition label but may have several feasible corridors. Counting every corridor candidate as a full observation would give ambiguous segments more physical-layer mass than network-layer mass. CLASP instead allocates one unit of mass across the candidate set and aggregates both layers over the same segments. Section III formalizes this construction and the resulting comparisons; Section IV describes the measurement inputs.

III. CLASP: THE PAIRED CROSS-LAYER AUDIT

A. Overview

CLASP compares two representations of the same observations. Within each country–measurement unit, every candidate-bearing segment contributes one unit of mass to a network-transition distribution and one unit, allocated among candidates, to a feasible-corridor distribution. Figure 1 places this paired construction at the center of the workflow: path extraction defines the observations, corridor projection supplies feasible support, and allocation and aggregation produce the distributions used for within-unit and cross-unit comparisons.

The inputs are traceroutes, probe metadata, IP geolocation and ASN mappings, AS context, and cable metadata. The main branch conditions both distributions on candidate-bearing segments. A separate trace-level branch reports exposure using all valid traceroutes and the presence of feasible inter-region candidates. Keeping these branches separate makes their different populations and denominators explicit.

B. Path Extraction

Country and measurement identity define the comparison unit before segments are aggregated. We group traceroutes by probe country and measured service, retaining source-side connectivity and observed destination selection. Hop countries remain segment attributes rather than grouping keys, so intermediate locations do not redefine the source population.

Each traceroute is normalized into its complete sequence of visible hops with locations, ASNs, and RTTs. Successive visible, geolocated hops form atomic segments. Each segment records endpoint locations, ASNs, countries, hop positions,

RTT difference, and bridged-hop count. Repeated hops and RTT samples retain their trace identity and ordering. When timeouts separate two visible hops, the bridged-hop count records the hidden gap for audit accounting.

When both endpoint ASNs are available, their ordered pair forms the segment’s network-transition label. A segment missing one endpoint ASN receives an ordered country-pair fallback label, which is strictly coarser. We record the fallback share for each unit. Audit eligibility requires that share to be at most 30%, together with at least 30 candidate-bearing segments, 10 probes, and 3 probe ASNs. These thresholds remove units whose network labels or observation support are too sparse for the paired summaries; they are quality floors rather than guarantees of representativeness.

C. Feasible Landing-Region Corridors

a) *Landing stations and cable lifecycle.*: For each segment endpoint, CLASP retrieves landing stations within 50 km and pairs them through cable systems in service on July 1, 2026. The catchment allows endpoint locations to be associated with nearby coastal facilities at the resolution of the geolocation input. Explicit cable path or branch information defines direct landing-point adjacency. When the inventory does not provide that information, the pipeline retains reachability at the metadata’s topology resolution and records that provenance state rather than presenting it as an observed cable traversal.

b) *Landing regions.*: Landing stations are grouped into regions with maximum pairwise distance 30 km. The diameter merges nearby facilities serving the same coastal direction while retaining separated coastal entry and exit patterns. A *landing-region corridor* is a direction-independent pair of regions connected by at least one feasible cable candidate. Several cable systems and exact landing pairs may support one corridor. Exact landing-pair, cable, and corridor identities remain linked in the outputs so that the aggregation can be audited at each resolution.

c) *Propagation feasibility.*: A candidate must satisfy geographic, lifecycle, and propagation constraints. Let D be the great-circle distance between its landing points, $v_f = 200$ km/ms the configured effective propagation speed in fiber, ΔRTT the segment’s RTT difference, and $\tau = 5$ ms the RTT tolerance. The candidate is propagation-feasible when

$$\Delta RTT + \tau \geq \frac{2D}{v_f}. \quad (1)$$

The inequality checks whether the observed round-trip increment can accommodate the minimum round-trip propagation time between the landing points. When ΔRTT is non-positive or inconclusive, CLASP retains the geographic and lifecycle constraints and flags the segment. The 200 km/ms speed and 5 ms tolerance are fixed configuration values in the reported analysis.

D. Candidate Allocation and Observation Mass

Candidate multiplicity must not increase a segment’s observation mass. Duplicate cable rows belonging to the same

CLASP One country-measurement unit; paired comparison over the same segments

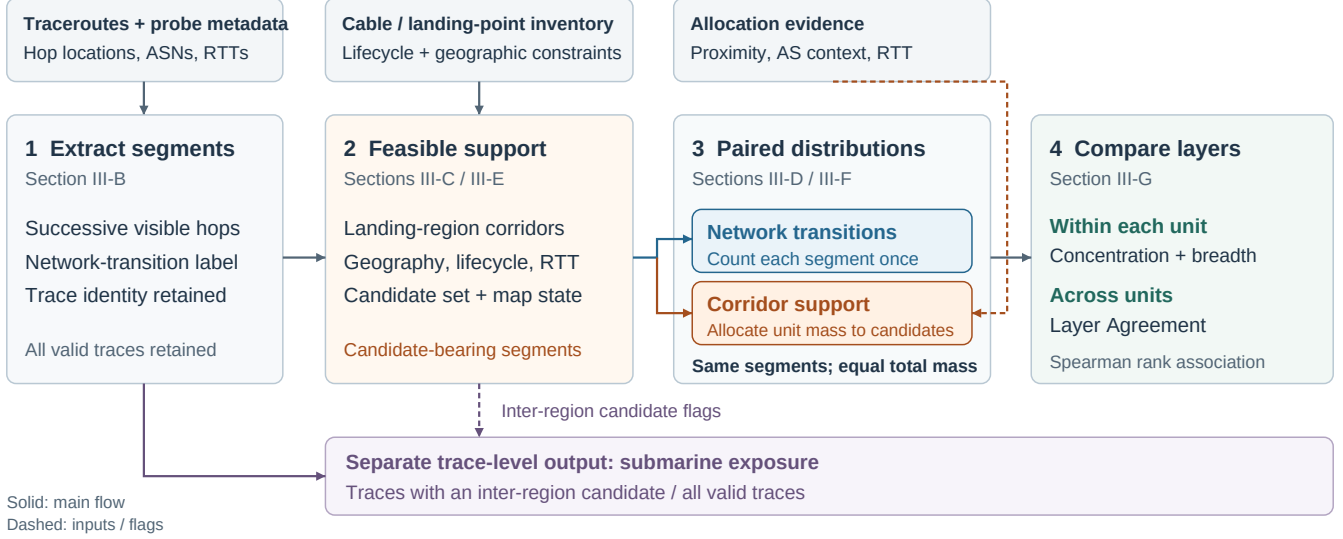


Fig. 1. CLASP preserves the same candidate-bearing segments and total observation mass in the two distributions. The main flow separates inputs, feasible support, paired construction, and comparison. The trace-level exposure branch uses all valid traces and inter-region candidate flags; dashed arrows denote supporting evidence or flags.

corridor are collapsed within an atomic segment. For a segment s with feasible corridor set $\mathcal{C}_s$, projection-score allocation distributes one unit of mass according to the candidates' relative support:

$$w_{s,c} = \frac{\sigma_{s,c}}{\sum_{c' \in \mathcal{C}_s} \sigma_{s,c'}}, \quad \sum_{c \in \mathcal{C}_s} w_{s,c} = 1, \quad (2)$$

where $\sigma_{s,c}$ sums the evidence-support scores of the cable candidates assigned to corridor c . The implementation forms each candidate score as the product of landing-proximity support, AS-economic support, RTT support, and a latency penalty. Landing-proximity support decreases with the distances from the two hop endpoints to their candidate landing points. AS-economic support uses endpoint-AS membership in cable-owner AS sets and relationships or owner-group reachability in the AS graph. RTT support compares the observed increment with the propagation lower bound and applies a penalty to highly inflated short-path increments; inconclusive increments receive neutral RTT factors. The candidate scores are summed by corridor and normalized as in Eq. (2), so every candidate-bearing segment contributes one unit of mass.

The three allocation rules retain different amounts of candidate-score information while preserving this unit-mass constraint. Projection-score weighting is the primary allocation for Layer Agreement and retains relative evidence support. Equal sharing assigns $w_{s,c} = 1/|\mathcal{C}_s|$ without using score magnitudes, and Top-1 assigns all mass to the highest-scoring corridor. Endpoint-AS information contributes both to network labels and to candidate ranking; equal sharing therefore provides a comparison that does not use the score magnitudes within the feasible sets. The rules express equal, graded,

TABLE I
SEGMENT-LEVEL PHYSICAL MAPPING RESOLUTION.

State	Operational definition
Single corridor	Feasibility inputs available and $ \mathcal{C}_s = 1$
Bounded multi-corridor	Feasibility inputs available and $ \mathcal{C}_s > 1$
No feasible corridor	Feasibility inputs available and $ \mathcal{C}_s = 0$
Insufficiently resolved	Insufficient location, path-visibility, or meta-data fields for stable corridor construction

and single- candidate support, not confirmed cable usage probabilities.

E. Mapping Resolution and Pipeline Accounting

Mapping states determine which segments enter the paired distributions. Table I distinguishes input coverage and candidate- set size. Single- and bounded multi-corridor segments enter the physical distribution and its paired network distribution. Segments with no feasible corridor or insufficient inputs remain in pipeline accounting; every valid traceroute remains in the exposure denominator.

Candidate-set size and aggregate concentration describe different levels. A unit may contain many bounded multi-corridor segments yet concentrate after their allocated mass repeatedly supports the same corridors. CLASP therefore retains the resolution state instead of using candidate multiplicity as a substitute for distribution breadth.

F. Paired Distributions

For an auditable country-measurement unit u , let S_u denote its candidate-bearing atomic segments and $\ell(s)$ the ordered

network label of segment s . The network-transition probability of label t is

$$p_u^N(t) = \frac{\sum_{s \in S_u} \mathbf{1}[\ell(s) = t]}{|S_u|}. \quad (3)$$

The corridor distribution sums the allocated masses in Eq. (2) and normalizes over the same S_u . Before normalization, each layer has total mass $|S_u|$: candidate ambiguity changes the allocation, not the amount contributed by a segment. The paired distributions thus compare how a fixed observation population is organized. Network labels preserve direction, whereas corridors are direction-independent coastal connections; the comparison retains this difference in category meaning.

Submarine exposure uses the complete valid-trace population. A traceroute is exposed when at least one of its segments has a feasible inter-region corridor. The number of exposed traces divided by all valid traces in the unit gives the exposure rate. This denominator differs intentionally from $|S_u|$, which conditions the distributions on candidate-bearing segments.

G. Distribution Summaries and Layer Agreement

Within-unit comparisons measure concentration and breadth separately. Top-2 share sums the two largest probabilities, recording mass in the dominant categories. Effective category count, $\exp[-\sum_i p_i \log p_i]$, expresses breadth as an equally weighted category count. Normalized entropy divides Shannon entropy by the logarithm of observed support size and describes evenness relative to that support. For each metric, network-to-corridor differences are computed within the same unit.

Layer Agreement is the Spearman rank association, across a reported set of units, between network Top-2 share and corridor Top-2 share. A coefficient approaching +1 indicates closely aligned concentration rankings; a coefficient near zero indicates little monotonic rank association, and a negative coefficient indicates opposing rankings. This comparison concerns ordering across units. Within-unit Top-2 differences and effective category counts separately measure how concentration and breadth change across layers. A group can therefore exhibit aligned rankings alongside substantial within-unit narrowing. We report Layer Agreement by measurement family, geography, and candidate-allocation rule.

For the frozen equal-share result, we quantify geographic uncertainty at the level where geography is assigned. A cluster-bootstrap replicate samples countries with replacement within each geographic group and carries every country–measurement unit from each sampled country together. We use 50,000 replicates and report percentile intervals. A separate two-sided reference test permutes the geographic labels of whole countries 100,000 times while retaining the observed number of countries in each group. Both procedures use seed 20260919. This country-clustered analysis is applied to the frozen equal-share unit table; projection-score and Top-1 results provide the allocation-robustness comparison.

TABLE II
CLASP MEASUREMENT CONFIGURATION.

Parameter	Value
Endpoint landing catchment	50 km
Landing-region maximum diameter	30 km
Fiber propagation speed	200 km/ms
RTT tolerance	5 ms
Same-city threshold	25 km
Cable lifecycle date	July 1, 2026
Inconclusive RTT	Retained and flagged
Timeout-bridged segment	Retained and flagged
Path scope	Complete visible sequence

A common 80% Top-2 threshold supplies four descriptive classes: broad at both layers, concentrated at both, network-broad/corridor-concentrated, and network-concentrated/corridor-broad. Continuous Top-2 shifts, effective counts, and normalized entropy accompany the classes so that threshold crossings do not replace the underlying distributions.

IV. DATASETS

The corpus retains source and target identity for the paired audit. All inputs are aligned to July 1, 2026 (UTC): network-path observations are combined with IP and AS annotations and cable lifecycle records for the analysis date. The following datasets supply the path population, the annotations used in projection, and the geographic grouping used in the comparisons.

A. RIPE Atlas Measurements

Our network observations comprise 18 public RIPE Atlas IPv4 traceroute measurements collected from 00:00 to 01:00 UTC. Thirteen measurements cover independently operated anycast DNS Root services. Wikipedia, Reddit, and `assets.nflxext.com` provide three application-facing observations. Measurements 5051 and 5151 use dynamic targets as multi-target UDP and ICMP topology references. Table III summarizes the corpus after canonical trace-identity deduplication.

Each record retains the destination address returned by RIPE Atlas. This preserves the multi-target structure of the two dynamic-target measurements and the destination instances selected by service measurements during the aligned window.

The two dynamic-target measurements provide a broader-target reference for the service populations. They sample many destinations during the window, whereas service measurements retain the destinations selected for each service. The comparison describes these observed populations; their different target-selection processes do not constitute a controlled test of service deployment or target-selection effects.

B. Supporting Datasets

Table IV lists the supporting datasets and versions. RIPE Atlas probe metadata supplies the probe country and source-network context [16]. IPinfo Location and ASN databases annotate visible hops with geographic and AS information [17].

TABLE III
RIPE ATLAS MEASUREMENT CORPUS.

Group	Target	MSM ID(s)	Raw	Valid
DNS Roots	A-M Root	5009, 5010, 5011, 5012, 5013, 5004, 5014, 5015, 5005, 5016, 5001, 5008, 5006	534,276	317,873
Applications	Wikipedia; Reddit; Netflix Assets	86710103; 176906957; 176517335	115,596	115,596
Topology references	IPv4 UDP; IPv4 ICMP	5051; 5151	57,442	57,442
Total	18 measurements	-	707,314	490,911

The Submarine Cable Map provides the cable and landing-point inventory, including lifecycle and organizational fields [1]. The analysis includes systems in service on the measurement date. CAIDA AS Relationships and the derived owner-AS associations supply supplementary ranking context [18]. A versioned country taxonomy classifies probe countries as island or archipelagic, landlocked, or other coastal.

V. RESULTS

The paired audit reveals cross-layer differences within country-measurement units and geographic variation in the association between their concentration rankings. We first quantify within-unit changes, then compare Layer Agreement across geographic groups and candidate-allocation rules. Exposure, absolute corridor breadth, and measurement-family summaries provide context for these results.

The processing pipeline retains 490,911 valid traceroutes from 707,314 raw records and yields 3,291,063 atomic segments. Of 2,432,559 mappable segments, 555,119 reach candidate landing points, 171,232 satisfy the corridor constraints, and 124,350 support at least one feasible inter-region corridor. The equal-share audit records 71,990 single-corridor and 52,360 bounded multi-corridor segments. After the eligibility filters in Section III-B, the analysis contains 370 auditable country-measurement units: 222 DNS Root, 53 application, and 95 topology-reference units.

A. Network and Corridor Diversity Differ within Units

Under equal-share allocation, 81 of 275 service-facing units (29.5%) have broad network support but concentrated feasible-corridor support. Using the 80% Top-2 threshold, this network-broad/corridor-concentrated class is one of four cross-layer outcomes (Fig. 5): 158 units (57.5%) remain broad at both layers, 18 (6.5%) are concentrated at both, and 18 (6.5%) are network-concentrated/corridor-broad. Four of 95 topology-reference units (4.2%) enter the first class; the other 91 remain broad at both layers.

Continuous measures reveal narrowing within as well as across concentration classes. Across all service-facing units, corridor Top-2 share increases in 184 of 275 (66.9%), and effective category count contracts in 204 (74.2%). Of 239 units with a broad network distribution, 178 (74.5%) move upward in Top-2 share: 81 cross the threshold, while 97 remain broad at both layers with a median increase of 6.3 percentage points. Thus, remaining in the broad-broad class can coexist with an increase in concentration. DNS Roots and applications enter

the network-broad/corridor-concentrated class at similar rates: 66 of 222 DNS units (29.7%) and 15 of 53 application units (28.3%). Their median continuous Top-2 shifts are 11.0 and 21.0 percentage points.

Singapore-Netflix remains in the broad-broad class, yet its corridor distribution is more concentrated: Top-2 share rises from 32.5% at the network layer to 65.0% at the corridor layer, while effective category count contracts from 19.86 to 3.41. By contrast, **New Zealand-Netflix** becomes less concentrated: Top-2 share falls from 63.6% to 37.6%, with effective count expanding from 5.18 to 6.92. **Singapore-H-Root** illustrates a different distinction, between absolute breadth and support-relative evenness: Top-2 share rises from 39.6% to 50.1%, and effective count contracts from 20.67 to 5.99 while normalized entropy changes only slightly.

Bounded multi-corridor segments account for 99.9%, 78.1%, and 98.2% of candidate-bearing observations in these three cases. Repeated overlap among the bounded sets produces different aggregate distributions under the same projection and aggregation rules. Together, the cases and four-class counts establish cross-layer heterogeneity and show why the audit reports both continuous measures and categorical outcomes.

B. Layer Agreement Differs by Geography

Under projection-score weighting, network and corridor concentration rankings correspond more closely among island and archipelagic service-facing units than among coastal-mainland units. Layer Agreement is 0.632 across 67 island and archipelagic units and 0.001 across 202 coastal-mainland units (Table V). These units represent 10 and 28 countries, respectively, and the observed coefficient difference is 0.631. The island and archipelagic group exhibits a stronger positive concentration-rank association; the coastal-mainland group exhibits almost no monotonic rank association. Landlocked units form a separate geographic category.

C. Geographic Ordering across Candidate Allocations

The geographic contrast retains the same direction under all three candidate-allocation rules. As allocation moves from equal shares to projection-score weighting and then Top-1, the island coefficient increases from 0.380 to 0.632 and 0.820, while the coastal-mainland coefficient changes from -0.081 to 0.001 and 0.105. The between-group difference consequently increases from 0.461 to 0.631 and 0.715. The equal-share result contains three additional coastal-mainland DNS units

TABLE IV
SUPPORTING DATASETS.

Dataset	Version	Content
RIPE Atlas probe metadata	2026-07-01	Probe country and source ASN
IPinfo Location MMDB	2026-07-01	Hop geolocation
IPinfo ASN MMDB	2026-07-01	IP-AS mapping
Submarine Cable Map	2026-07-01	694 cables; 1,916 landing points
CAIDA AS Relationships	2026-07-01 release	AS relationships
Owner-AS mapping	2026-07-01 inputs	Owner-AS associations
Country geography taxonomy	2026-07-01	Operational geography classes

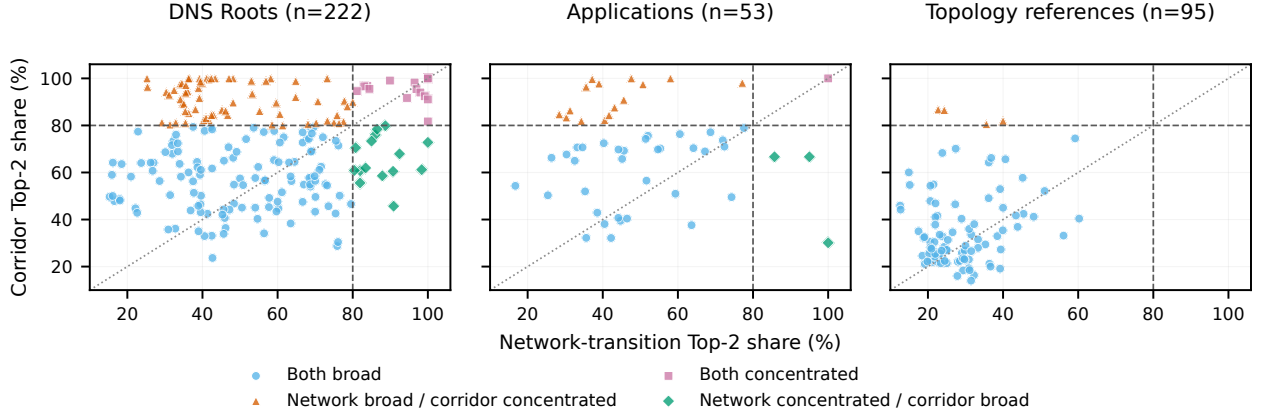


Fig. 5. Equal-share network-transition and corridor Top-2 shares for 370 auditable units. The 80% lines define the descriptive four-class view; the diagonal separates increases from decreases in concentration.

TABLE V
SERVICE-FACING LAYER AGREEMENT. PARENTHESES GIVE UNIT COUNTS.

Allocation	Island/arch.	Coastal mainland	Difference
Equal share	0.380 (67)	-0.081 (205)	0.461
Projection score	0.632 (67)	0.001 (202)	0.631
Top-1	0.820 (67)	0.105 (202)	0.715

relative to the projection-score and Top-1 comparisons. Across these allocation-specific cohorts, the island coefficient exceeds the coastal-mainland coefficient under equal, graded, and Top-1 support. Equal sharing preserves this ordering while assigning corridor mass independently of score magnitudes.

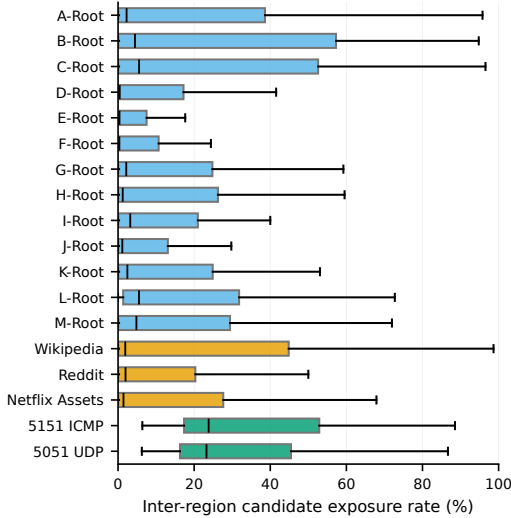
The reproducible equal-share result supports a country-level uncertainty check. Its 67 island and archipelagic units represent 10 countries, and its 205 coastal-mainland units represent 30 countries. Country-cluster bootstrap 95% intervals are $[-0.203, 0.760]$ for the island coefficient, $[-0.284, 0.238]$ for the coastal-mainland coefficient, and $[-0.213, 0.902]$ for their 0.461 difference. A two-sided country-label permutation gives $p = 0.161$. The wide interval for the difference includes zero. This check quantifies uncertainty across 10 island and 30 coastal country clusters; the three allocation rules separately compare the observed ordering under alternative candidate-mass assignments.

D. Inter-Region Candidate Exposure

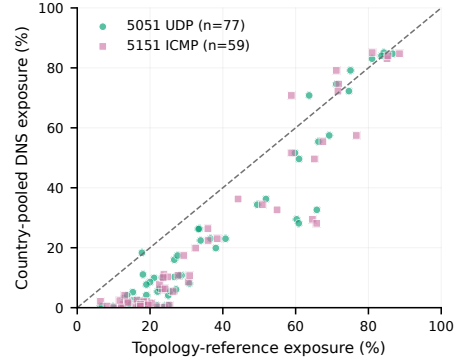
Inter-region candidates occur infrequently for most service-facing populations when measured over all valid traces. The DNS family contains 945 country-measurement units with at least 30 valid traceroutes across 77 countries. Pooling eligible Root observations within each country by valid-trace count yields median DNS exposure of 5.50% (IQR 1.10–23.04%; mean 17.80%). The difference between the median and mean records a pronounced upper tail across countries.

The three application measurements occupy the same low-exposure range. Median country-measurement exposure is 1.90% for Wikipedia, 1.96% for Reddit, and 1.43% for Netflix Assets. The dynamic multi-target topology references reach 23.24% for MSM 5051 and 23.81% for MSM 5151. These figures describe how often paths in each observed population contain a feasible inter-region corridor; they do not describe concentration after entry.

Country-matched comparisons describe exposure differences between service and topology-reference populations with the same source-country identity. DNS exposure is lower than MSM 5051 in 70 of 77 shared countries, with a median paired difference of -14.52 percentage points. It is lower than MSM 5151 in 54 of 59 shared countries, with a median difference of -14.82 points. The paired direction holds for 90.9% and 91.5% of shared countries, respectively (Fig. 2).



(a) Country-measurement exposure.



(b) Country-matched comparison.

Fig. 2. Inter-region candidate exposure in the observed measurement populations.

E. Absolute Exposure and Breadth by Geography

The geographic groups also differ in how often DNS paths enter the candidate space. Giving each country equal weight, island and archipelagic countries have median country-pooled DNS exposure of 49.61% across 9 countries, compared with 7.88% for 54 other coastal countries and 1.70% for 14 landlocked countries.

Conditional equal-share corridor breadth follows a different ordering. Island and archipelagic countries have median corridor Top-2 share of 53.3% and 5.58 effective corridors. Other coastal countries reach 75.4% and 3.39. The two represented landlocked countries reach 80.2% and 2.51. Across 38 countries with both pooled exposure and an auditable corridor summary, exposure has Spearman association -0.29 with corridor Top-2 share and 0.30 with effective corridor count. These quantities show that frequent entry and conditional concentration are not interchangeable.

F. Layer Agreement within Measurement Families

Within each measurement family, network and corridor concentration rankings show weak association (Table VI). Under projection-score weighting, Layer Agreement is 0.176 for 219 DNS units, 0.229 for 53 application units, and 0.012 for 95 topology-reference units, all smaller than the island and archipelagic coefficient. Equal-share and Top-1 allocation change all three coefficients, with the largest change occurring for DNS Roots. The equal-share audit contains 222 DNS units, whereas the family allocation comparison contains 219; the tables retain these allocation-specific counts.

Absolute equal-share corridor breadth also differs among the observed families. Table VII reports network and corridor medians. DNS Roots and applications have median corridor Top-2 shares of 72.0% and 70.3%, compared with 30.6% for topology references. Their median effective corridor counts are 3.84 and 3.51, compared with 18.01. The two topology-

TABLE VI
LAYER AGREEMENT BY MEASUREMENT FAMILY.

Family	n	Equal	Score	Top-1
DNS Roots	219	0.096	0.176	0.330
Applications	53	0.030	0.229	0.372
Topology references	95	0.139	0.012	0.122

reference measurements deliberately sample broader target populations, providing a breadth reference for the service-facing distributions.

The high-concentration tail follows the same descriptive family ordering. The leading two corridors receive at least 80% of mass in 83 of 222 DNS units (37.4%), 16 of 53 application units (30.2%), and 4 of 95 topology-reference units (4.2%). Wikipedia, Reddit, and Netflix Assets have median corridor Top-2 shares of 68.3%, 71.0%, and 71.6%, with 4.25, 3.25, and 3.47 effective corridors. Their median Top-2 shifts are 11.0, 25.6, and 23.6 percentage points, respectively.

Normalized entropy supplies a support-relative view. Median entropy reduction is 0.126 for DNS Roots, 0.164 for applications, and 0.081 for topology references. The family summaries therefore agree on observed absolute breadth and concentration, while the Layer Agreement coefficients show that within-family rankings remain weak and allocation-sensitive.

VI. ROBUSTNESS, SCOPE, AND IMPLICATIONS

Interpreting the paired audit requires distinguishing measured cross-layer differences from the conditions under which they were obtained. Section V-C reports the allocation comparison and country-clustered equal-share uncertainty check. Here we examine landing-region resolution, identify the geographic and population scope, and explain what the two levels of comparison contribute to diversity measurement.

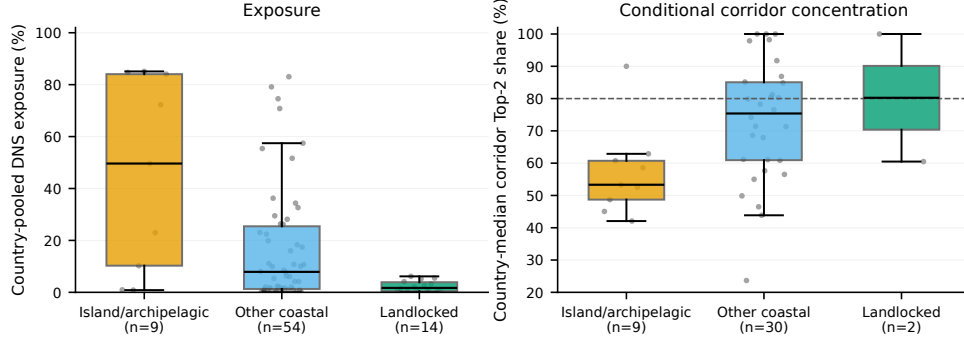


Fig. 3. DNS exposure and equal-share corridor concentration by geography. Country counts refer to the DNS country-level summary and are distinct from the service-facing unit counts in Table V.

TABLE VII
EQUAL-SHARE NETWORK-TRANSITION AND CORRIDOR DISTRIBUTIONS.

Family	Units	Net. Top-2	Corr. Top-2	Δ Top-2	Eff. network	Eff. corridor
DNS Roots	222	53.1%	72.0%	+11.0 pp	7.45	3.84
Applications	53	45.2%	70.3%	+21.0 pp	9.89	3.51
Topology references	95	27.3%	30.6%	+2.7 pp	24.34	18.01

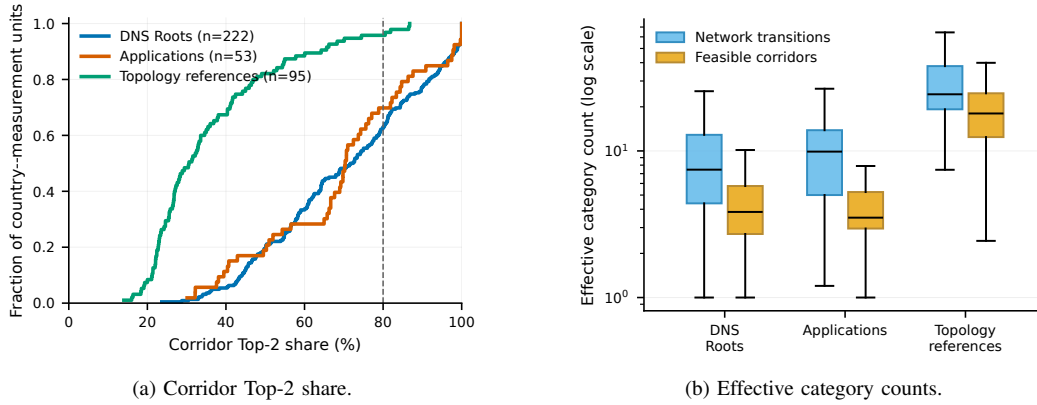


Fig. 4. Equal-share corridor concentration and breadth by measurement family. The families contain different target populations.

A. Landing-Region Resolution

The A-Root landing-region analysis varies maximum diameter over 10, 20, 30, 40, and 50 km while holding the endpoint catchment at 50 km and the RTT tolerance at 5 ms. Table VIII and Fig. 6 report the auditable-unit count, single-corridor share, and median Top-2 share. Exact landing-pair and cable candidate sets have mean Jaccard 1.0 on segments shared with the 30-km baseline. The sweep records changes in region grouping, the candidate-bearing population, and auditable-unit counts within A-Root.

The sweep shows a clear resolution transition. The 10–30-km settings retain a finer coastal partition, whereas single-corridor share rises to 90.1% and 94.8% at 40 and 50 km and median Top-2 share reaches 81.6% and 84.9%. We use 30 km because it remains on the fine side of this observed transition while consolidating nearby facilities; 40–50 km merge substantially more candidate support into single corridors. The

TABLE VIII
A-ROOT LANDING-REGION DIAMETER SENSITIVITY.

Diameter	Single-corridor	Median Top-2	Units
10 km	47.4%	53.6%	16
20 km	46.2%	62.8%	15
30 km	54.8%	64.4%	15
40 km	90.1%	81.6%	13
50 km	94.8%	84.9%	13

A-Root check therefore supplies an empirical basis for the reporting resolution within the tested A-Root subset and an operational basis for the 30-km setting used throughout the audit.

B. Dependence on Geolocation and Candidate Construction

Every projection begins with IPinfo hop geolocation and the fixed 50-km endpoint catchment. Geographic, lifecycle,

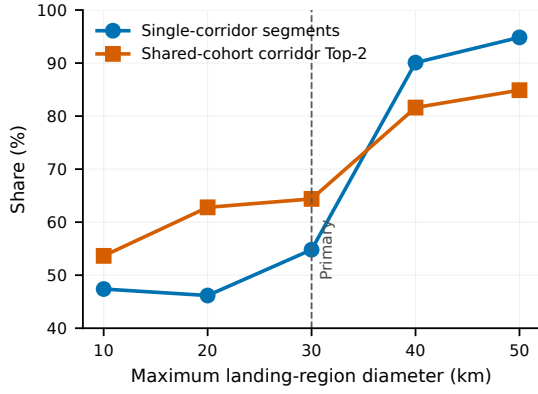


Fig. 6. A-Root landing-region diameter sensitivity under a fixed 50-km endpoint catchment and 5-ms RTT tolerance.

and propagation constraints then construct the feasible candidate set. Cable lifecycle filtering aligns the inventory with July 1, 2026; a recorded inventory refresh changed 222 of 694 cable records, including 26 planned-flag changes and 47 revised service dates. The output retains exact landing-pair, cable, corridor, and mapping-resolution identities, making each projected distribution traceable to its geolocation and inventory inputs.

C. Snapshot and Population Scope

All 18 measurements cover 00:00–01:00 UTC on July 1, 2026, and the supporting datasets are aligned to the measurement date. This common observation window supports cross-sectional comparisons among services and geographic groups. The reported distributions describe the paths observed during that interval.

Reported quantities are observation-weighted over participating RIPE Atlas probes and therefore represent the probes and source ASNs present in each country-measurement population. The eligibility thresholds of 10 probes and 3 probe ASNs remove sparse units from the paired analysis.

The service measurements retain their selected instances, while the dynamic multi-target references deliberately sample many destinations during the same window. The reference distributions therefore provide a broader-target context for the service-facing populations. Anycast observations jointly reflect deployed footprint, BGP selection, interconnection, and source- network context.

D. Implications for Diversity Measurement

A network-layer diversity summary has two distinct interpretations: it can describe the breadth of a particular population, or rank that population against others. The within-unit comparison measures how the first description changes at the corridor layer; Layer Agreement measures the correspondence between the two rankings. Stronger rank association can coexist with substantial within-unit narrowing. Reporting both therefore identifies what is preserved across layers without treating ranking agreement as equality of physical breadth.

The DNS country-level summaries put the geographic rank contrast in context. Island and archipelagic countries have a lower median corridor Top-2 share than other coastal countries (53.3% versus 75.4%) and a higher median effective corridor count (5.58 versus 3.39). Thus, the observed DNS country-level medians show broader corridor support for the island group. Across 38 countries with both exposure and corridor summaries, exposure has Spearman associations of -0.29 with corridor Top-2 share and 0.30 with effective corridor count. These statistics describe entry frequency and conditional breadth at the country level, complementing the service-facing unit-level Layer Agreement comparison.

A possible explanation for the geographic contrast is that maritime access and network organization impose partly shared constraints on concentration rankings for island and archipelagic sources. In coastal-mainland settings, terrestrial routing, domestic backhaul, and interconnection may allow network concentration to vary more independently of feasible submarine corridors. This is a mechanism hypothesis; the present comparisons measure the geographic association. Distinguishing these explanations would require evidence linking the relevant access and routing structures to individual country-measurement outcomes.

For diversity audits, the unit-level results provide concrete inspection targets: 81 service-facing populations have broad network support but concentrated corridor support, and the continuous summaries identify additional narrowing among the 158 populations classified as broad at both layers. The geographic comparison adds context to interpreting network-layer rankings: they track corridor concentration more closely in the observed island and archipelagic group than in the coastal-mainland group. Together, these results support checking physical-corridor concentration alongside network diversity when evaluating service paths.

Corridor concentration can be combined with cable multiplicity within a corridor, landing-station redundancy, shared terrestrial backhaul, correlated hazards, available rerouting capacity, and repair processes in a broader resilience assessment. CLASP supplies the cross-layer concentration component and preserves the exact candidate identities needed for that follow-on analysis.

E. Ethics and Reproducibility

The study uses archived public RIPE Atlas measurements and infrastructure metadata and reports aggregate country-measurement statistics. It does not target individual users or report individual probe addresses. Candidate assignments are feasible hypotheses and should not be interpreted as disclosure of an operator's actual cable use.

The released measurement IDs, configuration values, mapping-resolution states, pipeline counts, and scripts reconstruct the primary tables and figures from versioned inputs. The paper's 80% four-class rule is the primary descriptive classification; the repository's tier-based auxiliary classification uses a different rule and must remain separately named in the artifact.

VII. CONCLUSION

CLASP measures how service-path diversity changes between network transitions and feasible physical corridors by comparing the same traceroute segments at both layers. In the aligned one-hour RIPE Atlas snapshot, 81 of 275 service-facing units have broad network support and concentrated corridor support under equal-share allocation, while effective category count contracts in 204 units (74.2%). Across units, projection-score Layer Agreement is 0.632 for 67 island and archipelagic units and 0.001 for 202 coastal-mainland units. The geographic ordering persists under equal-share and Top-1 allocation, with between-group differences of 0.461, 0.631, and 0.715 across the three rules. These results describe two distinct properties: changes within a population and correspondence between population rankings. A cross-layer diversity audit needs both, because similar ordering does not imply similar physical breadth.

VIII. USE OF AI DISCLOSURE

OpenAI ChatGPT and Codex assisted with code development, plotting scripts and language revision. Their contribution was limited to these tasks, and the authors validated the outputs against the source data and analysis results.

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